Unimodality of nested-box differences and near-full q-rook polynomials

Aurantium

13 October 2026

Introduction

For every integer $\ell \geq 1$ and partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_\ell > 0)$, let

$$B_\lambda = \{(i, j) : 1 \leq i \leq \ell,\ 1 \leq j \leq \lambda_i\}.$$

A placement is a set of cells with at most one cell in each row and column. A rook (a, b) cancels the cells (i, b) with $i \leq a$ and (a, j) with $j \leq b$. For a placement w, let $\mathrm{inv}(w)$ count the cells not cancelled by any rook, and set

$$R_r(\lambda; q) = \sum_{\substack{w \subseteq B_\lambda \text{ a placement} \\ |w| = r}} q^{\mathrm{inv}(w)} \in \mathbb{Z}_{\geq 0}[q].$$

Here q is an indeterminate. A coefficient sequence is unimodal when it weakly increases and then weakly decreases; zero polynomials and zero padding are allowed. The target is

$$\boxed{R_{\ell-1}(\lambda; q) \text{ is unimodal for every such } \lambda.}$$

This is the final question in Lewis–Morales, arXiv:2410.19714v3, Remark 6 (page 5); their rook conventions are on pages 1–2. The setting is finite integer combinatorics. No field, characteristic, regularity, or asymptotic hypothesis is required.

Main result

The near-full polynomial is unimodal for every Ferrers board. The proof establishes a stronger statement about nested products of finite chains. It uses neither a Lefschetz property nor log-concavity of the rook polynomial, and does not depend on finite computations. The product-difference reduction and Hilbert-series interpretation originated in unpublished preliminary research notes; both are derived explicitly below.

Theorem 1 (Nested-box difference)**.** *Let $n \geq 1$ and $u, v \in \mathbb{Z}_{\geq 0}^n$ satisfy $v_i \leq u_i$ for every i. Put*

$$P_u(q) = \prod_{i=1}^{n} (1 + q + \cdots + q^{u_i}).$$

Then $P_u(q) - P_v(q)$ has nonnegative, unimodal coefficients.

The theorem includes $u = v$, where the difference is zero. Its application to near-full rook polynomials has $u_i = a_i$ and $v_i = a_i - 1$, after separating negative and zero a_i. The proof is entirely elementary; all auxiliary inequalities and the chain decomposition used below are proved here.

1. Checking the rook reduction and boundary cases

Write the row lengths increasingly as $b_1 \le \cdots \le b_\ell$ and set $a_i = b_i - i + 1$. Write $[m]_q = 1+q+\cdots+q^{m-1}$ for $m \ge 1$ and $[0]_q = 0$.

Proposition 1 (Near-full product difference)**.** *If some $a_i < 0$, then $R_{\ell-1}(\lambda; q) = 0$. Otherwise,*

$$R_{\ell-1}(\lambda; q) = \prod_{i=1}^{\ell} [a_i + 1]_q - \prod_{i=1}^{\ell} [a_i]_q. \tag{1}$$

Proof. If $a_i < 0$, the i shortest rows have at most $b_i \le i - 2$ columns available. A placement with only one omitted row must put at least $i - 1$ rooks into those rows, which is impossible.

Assume henceforth that every $a_i \ge 0$. Process rows from the shortest (bottom) row to the longest (top) row, and fix the unique omitted row i. Every column used earlier exists in the current row and has already been cancelled there. In a row with w remaining columns, placing a rook leaves $0, 1, \ldots, w - 1$ uncancelled cells, once each as the rook position varies. Its weight sum is $[w]_q$. No later rook can cancel a cell in this lower row. Thus choices may be summed row by row, since the number w is independent of the previous column choices.

Before row i, the available widths are a_j; after row i, they are $a_j + 1$. The omitted row itself contributes a_i uncancelled cells. Its contribution is therefore

$$q^{a_i} \prod_{j<i} [a_j]_q \prod_{j>i} [a_j + 1]_q.$$

A width zero gives a zero factor, correctly excluding that placement type. Summing over i, and using $[a_i + 1]_q - [a_i]_q = q^{a_i}$, gives the telescoping product identity (1). □

If some $a_i = 0$, the second product vanishes and the first is a product of q-integers. Such products are symmetric and unimodal, as also follows from the lemmas below. If all $a_i \ge 1$, Theorem 1 applies to $u = a$ and $v = a - (1, \ldots, 1)$. This covers every board and every ℓ. In particular, $\ell = 1$ gives $R_0(\lambda; q) = q^{\lambda_1}$.

The unrestricted positive-vector problem is not an artificial enlargement: any multiset of positive a_i can be sorted increasingly and realized by $b_i = a_i + i - 1$, then reversing the b_i to make a partition. The basic non-log-concave example remains compatible with the result: $a = (1, 3)$, corresponding to $\lambda = (4, 1)$, gives

$$R_1((4, 1); q) = q + q^2 + 2q^3 + q^4.$$

Its unimodality is not log-concavity, since $1^2 < 1 \cdot 2$ at degree 2.

2. Two elementary interval-sum inequalities

For a finitely supported sequence $h = (h_i)_{i \in \mathbb{Z}}$ of nonnegative real numbers, log-concavity means $h_i^2 \ge h_{i-1} h_{i+1}$ for all i. "No internal zeros" means its positive support is an integer interval.

Lemma 1 (Sliding and shrinking sums)**.** *Suppose h is log-concave with no internal zeros. For any integer S and integer $m \ge 0$, both sequences*

$$U_k = \sum_{i=k-m}^{k} h_i, \qquad V_k = \sum_{i=k-m}^{S-k} h_i$$

are log-concave, with an empty sum interpreted as zero.

Proof. Fix an integer t. The sequence $w_t(i) = h_i h_{t-i}$ is symmetric about $t/2$ and nondecreasing as i approaches $t/2$ from below. Indeed, on its positive support,

$$\frac{w_t(i+1)}{w_t(i)} = \frac{h_{i+1}/h_i}{h_{t-i}/h_{t-i-1}} \geq 1 \quad \text{if } 2i+1 \leq t,$$

because adjacent ratios of a positive log-concave sequence are nonincreasing. The zero extension preserves this conclusion. Consequently, a centered integer interval maximizes the sum of w_t among sets of the same cardinality; its sum also dominates every set of smaller cardinality.

Let $I = [L, U] \cap \mathbb{Z}$. In $(\sum_I h_i)^2$, the terms on the anti-diagonal $i + j = t$ sum over

$$J = I \cap (t - I),$$

a centered interval about $t/2$. For the product of the sums over $I - 1$ and $I + 1$, the corresponding set is exactly $J - 1$. The centered-interval comparison proves the sliding inequality after summing over all t.

For the shrinking inequality, the neighboring intervals are $[L-1, U+1]$ and $[L+1, U-1]$. Put $d = U - L + 1$. If $d \leq 2$, the latter interval is empty and the desired inequality is immediate; if the current interval is empty, its next shrinking interval is also empty. For $d \geq 3$, the anti-diagonal cardinalities in the square and in the neighboring product are, respectively,

$$|J| = \max\{0, \min(t - 2L + 1, 2U - t + 1, d)\},$$
$$|J'| = \max\{0, \min(t - 2L + 1, 2U - t + 1, d - 2)\}.$$

These formulas simply count pairs of integers in intervals of lengths d, d, or $d + 2, d - 2$, whose sum is t. Thus $|J'| \leq |J|$. Since J is centered, its w_t sum is at least the sum over J'. Summation over t proves $V_k^2 \geq V_{k-1} V_{k+1}$. □

3. Log-concavity of the lower-half first differences

For a nonnegative integer vector w (including the empty vector), set $T = \sum_i w_i$, and write $p_w(k) = [q^k] P_w(q)$, extended by zero outside $0 \leq k \leq T$. Define

$$g_w(k) = \begin{cases} p_w(k) - p_w(k-1), & 0 \leq k \leq \lfloor T/2 \rfloor, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

The truncation is part of the definition, not an assertion about the signs of the untruncated differences.

Lemma 2 (Difference profile)**.** *For every w, g_w is nonnegative and log-concave, and is positive exactly on an interval $\{0, 1, \ldots, K_w\}$. In particular $g_w(0) = 1$ and P_w is symmetric and unimodal. More explicitly, let $w = (z, m)$, $S = \sum z_i$, and $h = g_z$. Then for every integer $k \geq 0$,*

$$g_{(z,m)}(k) = \sum_{i=k-m}^{\min(k, S-k)} h_i. \tag{3}$$

If $\operatorname{supp}(h) = \{0, \ldots, r\}$*, its support endpoint is*

$$K_{(z,m)} = \min\left\{ m + r,\ S,\ \left\lfloor \frac{S+m}{2} \right\rfloor \right\}. \tag{4}$$

Proof. Induct on the number of coordinates. For the empty vector, $P = 1$ and $g(0) = 1$, with all other values zero. Each P_z is symmetric about $S/2$, since every factor is symmetric. Symmetry and telescoping give

$$p_z(k) = \sum_{i=0}^{\min(k,S-k)} h_i,$$

where a negative upper endpoint means zero. For $0 \le k \le \lfloor (S+m)/2 \rfloor$, multiplication by $1 + q + \cdots + q^m$ gives

$$g_{(z,m)}(k) = p_z(k) - p_z(k-m-1),$$

which is (3); here $k - m - 1 < S/2$ and is smaller than both k and $S - k$. For larger k, the interval in (3) is empty, so the same identity agrees with the stipulated truncation.

By induction, h is positive on $[0, r]$ and zero elsewhere, with $r \le \lfloor S/2 \rfloor$. The interval in (3) meets $[0, r]$ exactly when

$$k \le m + r, \qquad k \le S, \qquad 2k \le S + m.$$

This proves nonnegativity and (4).

To prove log-concavity, put $s = \lfloor S/2 \rfloor$. For $k \le s$, (3) equals the sliding sum U_k from Lemma 1. For $k \ge s$, it equals the shrinking sum V_k: when $S = 2s + 1$ and $k = s$, the only extra term of V_s is $h_{s+1} = 0$. Thus all log-concavity inequalities except possibly the one centered at s follow directly from that lemma. At s, we have

$$g_{(z,m)}(s-1) = U_{s-1}, \quad g_{(z,m)}(s) = U_s, \quad g_{(z,m)}(s+1) \le U_{s+1},$$

which proves the missing inequality as well. If $s = 0$, the inequality involving index -1 is automatic because that value is zero. The induction is complete. The last assertion follows from $g_w \ge 0$ and the symmetry of P_w. □

4. A shifted ratio that can cross a level only once

Lemma 3 (One-coordinate comparison)**.** *Let $w' = w + e_j$. For $k \ge 0$ on the positive support of g_w, the sequence*

$$\frac{g_{w'}(k+1)}{g_w(k)} \tag{5}$$

is nonincreasing in k. Moreover, if $g_w(k) = 0$ for $k \ge 0$, then $g_{w'}(k+1) = 0$.

Proof. Permute coordinates so the changed coordinate is last. Write $w = (z, m)$, $S = \sum z_i$, $h = g_z$, and $f_m = g_{(z,m)}$. By (4), $K_{m+1} \le K_m + 1$, proving the support assertion. For positive $f_m(k)$, the denominator and numerator of (5) are interval sums with the same lower endpoint $L = k - m$ and respective upper endpoints

$$\min(k, S-k) \quad \text{and} \quad \min(k+1, S-k-1).$$

There are three regimes.

If $2k \le S - 2$, the ratio is

$$1 + \frac{h_{k+1}}{\sum_{i=k-m}^{k} h_i} \ge 1. \tag{6}$$

Where $h_{k+1} > 0$, the added fraction is the reciprocal of

$$\sum_{j=1}^{m+1} \frac{h_{k+1-j}}{h_{k+1}}.$$

For a positive log-concave sequence, every backward ratio h_{t-j}/h_t is nondecreasing as t increases. This follows by expressing it as a product of the nondecreasing ratios h_{a-1}/h_a; terms below index zero are zero until they enter the support. Thus the displayed sum is nondecreasing, and (6) is nonincreasing. Once $h_{k+1} = 0$, it stays zero as k increases, and the ratio is 1.

If S is odd and $k = (S-1)/2$, the upper endpoints coincide and the ratio is exactly 1.

If $2k \geq S$, put $U = S - k$. The ratio is

$$1 - \frac{h_U}{\sum_{i=L}^{U} h_i} \leq 1. \tag{7}$$

On the positive denominator support, $U \geq 0$. If $h_U = 0$, necessarily $U > r$, so the ratio is initially 1. Once $h_U > 0$, the subtracted fraction is the reciprocal of

$$\sum_{j=0}^{U-L} \frac{h_{U-j}}{h_U}.$$

When k increases by one, U decreases by one, L increases by one, and the range of j shortens by two. Each surviving backward ratio weakly decreases by log-concavity. Hence this sum is nonincreasing, its reciprocal is nondecreasing, and (7) is nonincreasing.

The early ratios are at least 1 and the late ratios at most 1, with the possible odd-parity bridge equal to 1. This proves the comparison across the transitions too. If the numerator reaches zero, the initial support property prevents any later positive numerator. All divisions above were restricted to positive denominators. □

Lemma 4 (Coordinatewise comparison)**.** *Let $v \leq u$ coordinatewise and $\delta = \sum_i (u_i - v_i)$. Then, for $s \geq 0$ with $g_v(s) > 0$,*

$$F(s) = \frac{g_u(s+\delta)}{g_v(s)} \tag{8}$$

is nonincreasing in s. If $g_v(s) = 0$ and $s \geq 0$, then $g_u(s+\delta) = 0$.

Proof. Choose a path of δ unit coordinate increments $v = w^{(0)}, w^{(1)}, \ldots, w^{(\delta)} = u$. Repeated support inclusion from Lemma 3 shows that a positive endpoint $g_u(s+\delta)$ forces every $g_{w^{(a)}}(s+a)$ to be positive. On this initial interval,

$$F(s) = \prod_{a=0}^{\delta-1} \frac{g_{w^{(a+1)}}(s+a+1)}{g_{w^{(a)}}(s+a)}.$$

Every factor is nonnegative and nonincreasing by Lemma 3, so their product is nonincreasing. The numerator subsequently vanishes permanently by Lemma 2, giving a zero tail. Iterating support inclusion also proves the last assertion. If $\delta = 0$, the ratio is identically 1 wherever defined, so the statement remains valid. □

5. The upper-half coefficients of a box ideal

Rank the finite box $C(u) = \prod_i \{0, 1, \dots, u_i\}$ by $|x| = \sum_i x_i$; its total rank is $A = \sum_i u_i$. A lower ideal is a subset J closed under decreasing coordinates.

Lemma 5 (Upper-half decrease)**.** *For any lower ideal $J \subseteq C(u)$ and any integer $t > A/2$,*

$$|\{x \in J : |x| = t\}| \le |\{x \in J : |x| = t-1\}|.$$

Proof. We first construct a saturated symmetric chain decomposition of $C(u)$. In a rectangle $\{0, \dots, p\} \times \{0, \dots, m\}$, take the chain

$$(0,0), (1,0), \dots, (p,0), (p,1), \dots, (p,m).$$

If both $p, m > 0$, the remaining rectangle is $\{0, \dots, p-1\} \times \{1, \dots, m\}$. Translate its second coordinate down by one and repeat. This produces chains whose endpoint ranks sum to $p + m$. For a product of more chains, inductively take each symmetric chain of the earlier factors, whose endpoint ranks are r and $S - r$, and tensor it with the next chain. Apply the rectangle construction to its length $S - 2r$ and the new chain length. Shifting the ranks back by r makes every new chain symmetric about $(S + m)/2$. This proves the required decomposition by induction.

If such a chain contains an element of rank $t > A/2$, its initial rank is at most $A - t < t$, so this element has an immediate predecessor on the chain. A lower ideal contains that predecessor whenever it contains the element. Mapping each rank-t element of J to its predecessor is injective, since distinct chains are disjoint. This proves the inequality. □

6. Proof of the nested-box theorem

Proof of Theorem 1. The case $u = v$ is immediate. Otherwise put $A = \sum_i u_i$, $B = \sum_i v_i$, and $\delta = A - B > 0$. Since both products are symmetric in their respective degrees, reversal in degree A gives

$$q^A(P_u(q^{-1}) - P_v(q^{-1})) = P_u(q) - q^\delta P_v(q) =: H(q). \tag{9}$$

Write $c_i = u_i - v_i$. The polynomial H counts the lower ideal

$$J = C(u) \setminus \{x \in C(u) : x_i \ge c_i \text{ for every } i\}.$$

Indeed, the removed upper box is the translate $c + C(v)$ and has rank-generating polynomial $q^\delta P_v(q)$. Thus H is coefficientwise nonnegative, and Lemma 5 proves that its coefficients are nonincreasing past $A/2$.

It remains to check the first half. Let $h_t = [q^t]H(q)$ and $h_{-1} = 0$. For $0 \le t \le \lfloor A/2 \rfloor$, we have $t - \delta \le B/2$; hence (2) gives

$$h_t - h_{t-1} = g_u(t) - g_v(t - \delta). \tag{10}$$

For $t < \delta$, this is nonnegative. For $t \ge \delta$, set $s = t - \delta \ge 0$. Where $g_v(s) > 0$, the sign of (10) is that of $F(s) - 1$ from Lemma 4. Since F is nonincreasing, these signs cannot return to positive after becoming negative. If $g_v(s) = 0$, support inclusion gives $g_u(s + \delta) = 0$ too, so the difference is zero. Thus the first-half coefficients weakly increase and then weakly decrease, with no later increase. Appending the nonincreasing upper half from Lemma 5 proves that H is unimodal. Reversal preserves unimodality, including zero padding, and (9) proves the theorem. □

Together with Proposition 1 and Lemma 2, this proves the exact near-full target, with no restriction on ℓ or the board size.

7. Algebraic interpretation and limits of the conclusion

The same proof gives a characteristic-independent numerical corollary. For a field K, $u \in \mathbb{Z}^n_{\geq 0}$, and $0 \leq c_i \leq u_i$,

$$K[x_1, \ldots, x_n]/(x_1^{u_1+1}, \ldots, x_n^{u_n+1}, x_1^{c_1} \cdots x_n^{c_n})$$

has Hilbert series

$$P_u(q) - q^{\sum_i c_i} P_{u-c}(q),$$

and therefore has a unimodal Hilbert function. This follows by counting monomials, not by asserting a Lefschetz property. For the near-full positive-a family, $c_i = 1$ for every i.

This distinction is necessary: Migliore–Miró-Roig–Nagel, Theorem 4.3, give the non-WLP examples $K[x_1, \ldots, x_r]/(x_1^r, \ldots, x_r^r, x_1 \cdots x_r)$ for $r \geq 3$. They lie in the displayed family. Their failure of WLP is not a failure of the numerical unimodality proved here. No assertion is made that arbitrary-rank rook polynomials are unimodal; Lewis–Morales, Proposition 4, disproves that different statement.

8. Exact computational checks and reproduction

The accompanying `verify.py` uses only the Python standard library and exact integers. It checks the reduction by literal placements and tests the auxiliary lemmas. The following counts and reported passes come from archived computation records. The complete suite was not rerun for this edition; the archived records do not independently certify every historical execution.

Archived test family	Reported count
Coefficient products against direct monomial enumeration	341
All nonempty partitions with at most 15 cells; literal rook cancellation	683
Zero polynomials among these boards	358
Positive multisets of lengths 2–8, entries 1–12; near-full unimodality	125,957
Nonnegative multisets of lengths 1–8, entries 0–8; g-log-concavity and recursion	24,309
Distinct one-coordinate increments of those multisets; shifted ratio and support	102,960
Deterministic broad nested-box pairs, dimensions 1–24, outer entries 0–99	50,000
Additional large one-coordinate comparisons and g-log-concavity tests	10,000

The broad tests also check the composed ratio of Lemma 4 directly. They include dimensions, parities, zero coordinates, equality cases, and highly unequal inner and outer sizes. These finite checks are diagnostics only; they are not premises of any lemma or theorem.

To reproduce all tests, with assertions enabled, run from the package:

```
python verify.py --out-dir reproduced_results
```

The deterministic generator, seed, enumeration order, and exact coefficient computations are specified in the source. `results/verification.json` records counts and SHA-256 digests of canonical JSON-line record streams, including the coefficient arrays. `results/board_cases.json` contains every

literal board result; `results/examples.json` gives exact edge examples and the first 20 broad-test cases. Full larger streams can be regenerated exactly; they are not required to verify the proof. The report and supplied results correspond to the default 50,000-case run, not the optional shortened run.

9. Attribution and status

The proof was developed with AI assistance. This manuscript has not undergone independent expert peer review or formal proof checking. No exhaustive literature-priority claim is made for the nested-box theorem.

The product-difference reduction and monomial Hilbert-series interpretation were present in unpublished preliminary research notes. The argument developed here proves the interval-sum inequalities, lower-half difference-profile log-concavity, and shifted comparisons, and combines them with the explicitly constructed symmetric chain decomposition.

References

Joel Brewster Lewis and Alejandro H. Morales, *Garsia–Remmel q-rook numbers are not always unimodal*, arXiv:2410.19714v3. Definitions, pages 1–2; Example 3 and Proposition 4; Remark 6, page 5. `https://arxiv.org/abs/2410.19714v3`.

Juan C. Migliore, Rosa M. Miró-Roig, and Uwe Nagel, *Monomial ideals, almost complete intersections and the Weak Lefschetz Property*, arXiv:0811.1023; *Transactions of the American Mathematical Society* **363** (2011), 229–257. Theorem 4.3 supplies the WLP obstruction mentioned above. `https://arxiv.org/abs/0811.1023`.

Bek Chase and Filip Jonsson Kling, *The strong Lefschetz property for some monomial almost complete intersections*, arXiv:2507.18516v1 (arXiv abstract title: *Monomial Almost Complete Intersections and the Strong Lefschetz Property*). Relevant background: Definition 2.1, Theorem 3.1, Lemma 3.3, and Proposition 4.2. These are not invoked to prove the unrestricted target. `https://arxiv.org/abs/2507.18516v1`.

Unpublished preliminary research notes: starting product-difference reduction, Hilbert-series interpretation, and finite computations. The reduction and Hilbert-series interpretation are derived in full above.